

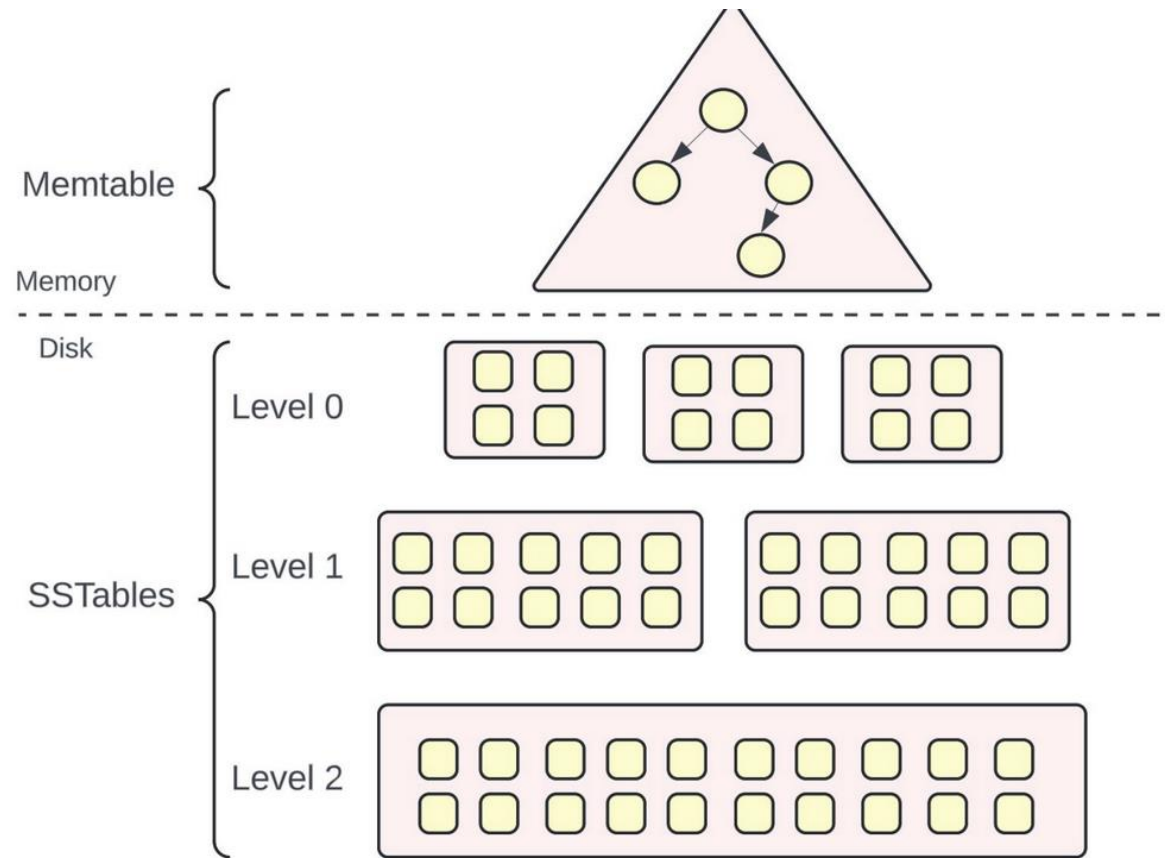
Learned LSM tree

Feb 18 2025

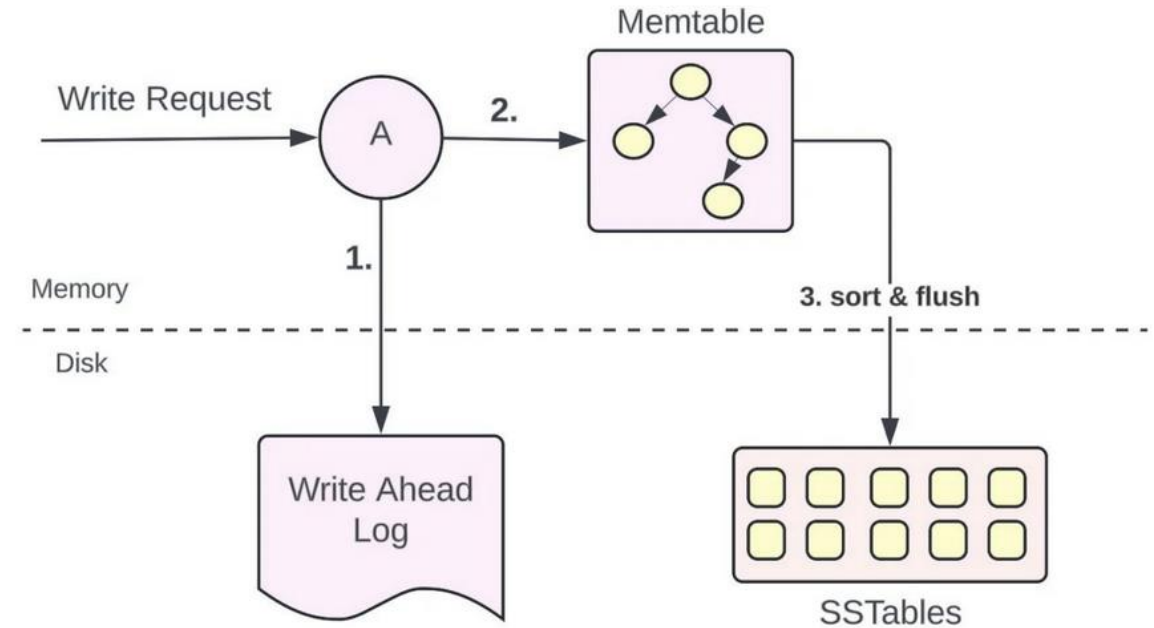
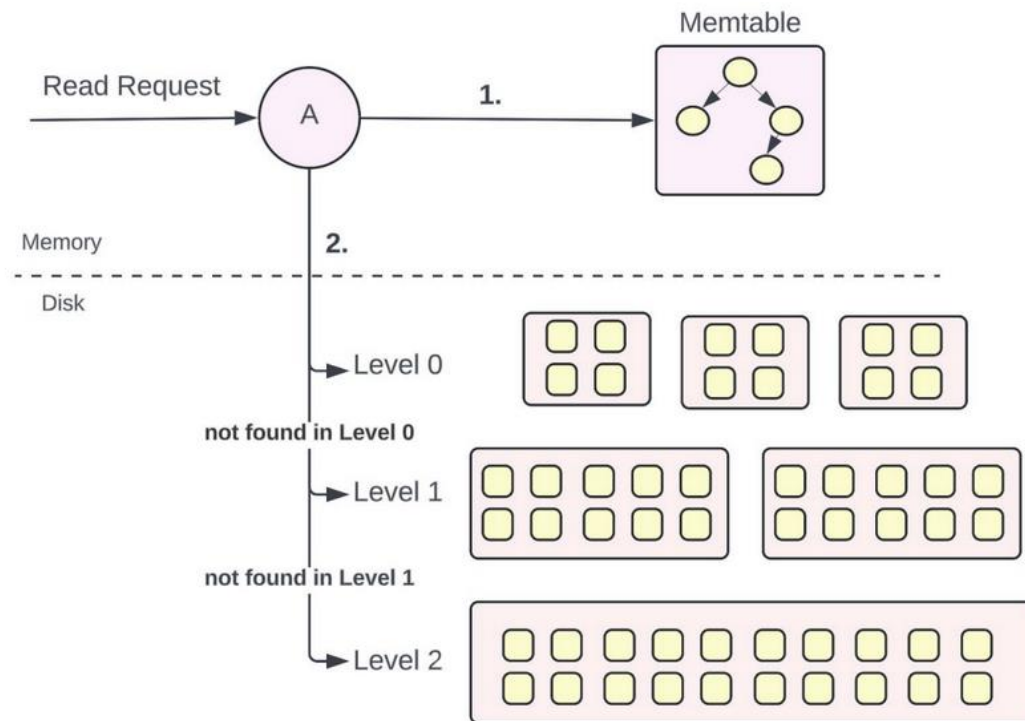
Learned index recap

- The main goal of a database index is _____
- B-tree stores object locations in a binary tree
- Learned B-tree uses hierarchy of models
 - Models learn CDF of keys
- Drawbacks of learned B-tree?

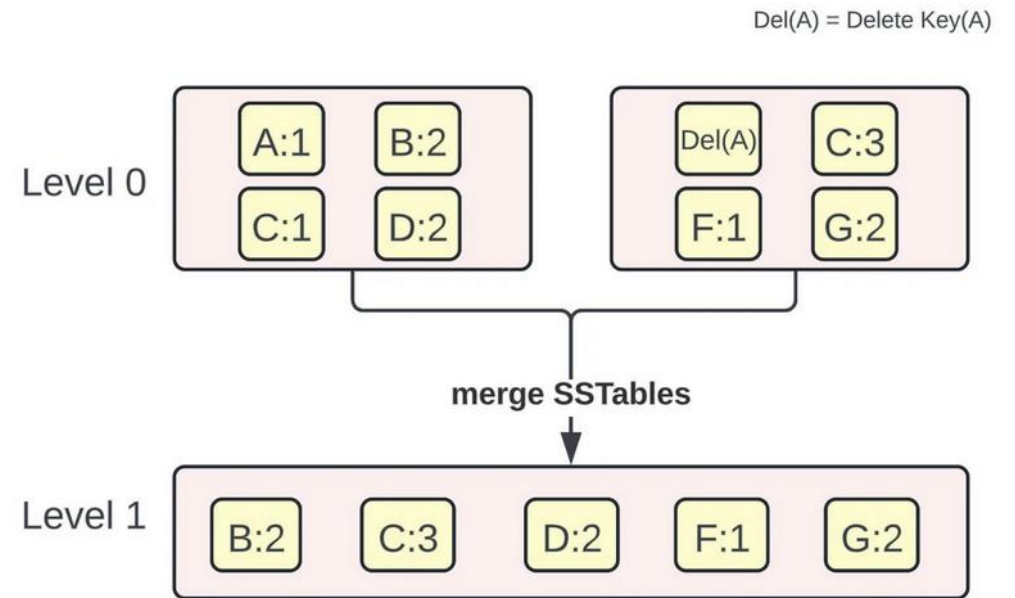
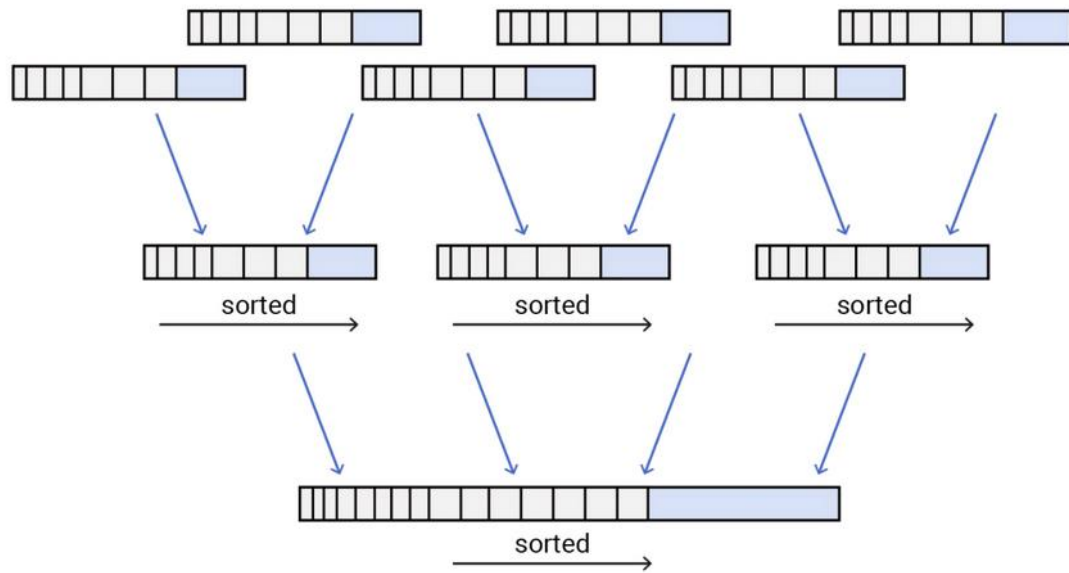
LSM tree structure



LSM reads and writes



LSM compaction



Analyzing LSM complexity

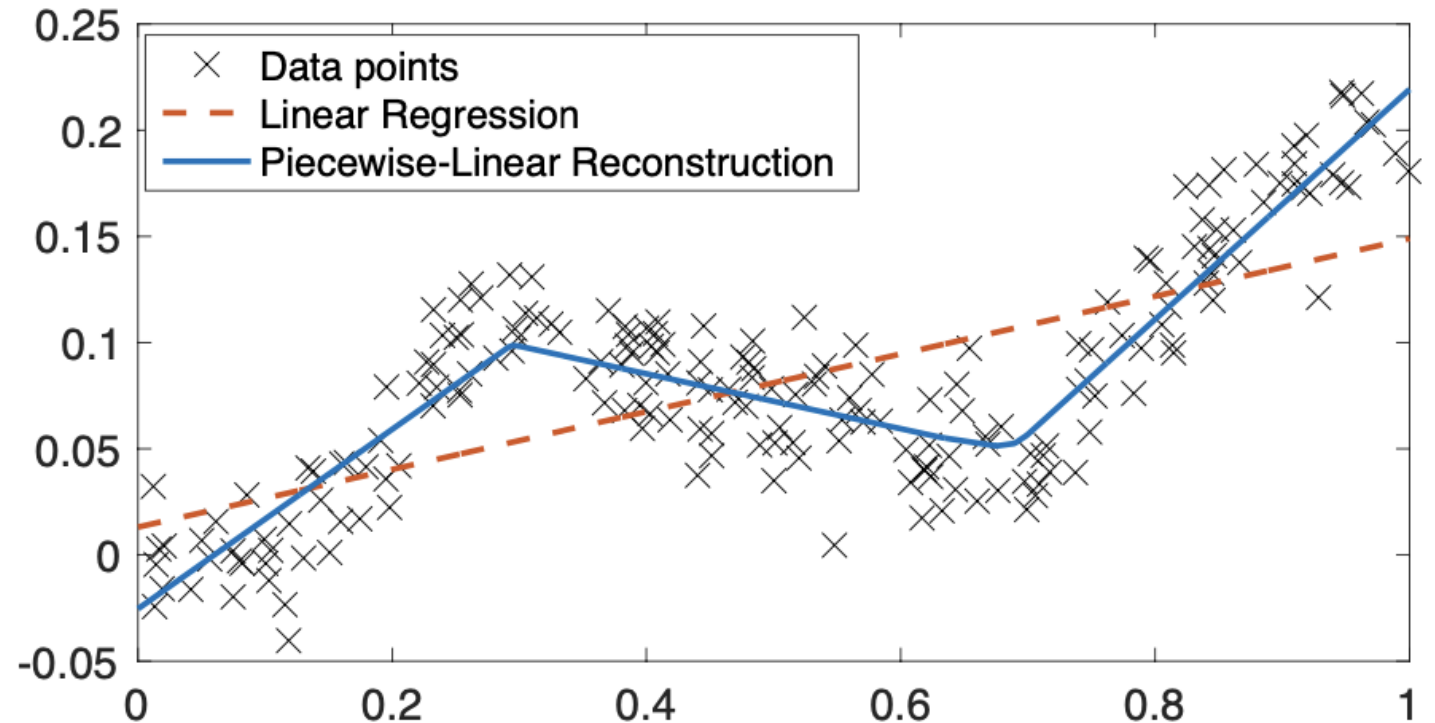
- Reading:
- Writing:
- Delete:

LSM tradeoffs and challenges

- Writes are cheap; reads are expensive
- Frequency of background compaction
 - Frequent: improves read performance; higher write amplification
 - Infrequent: poor read performance; lower write amplification
- Frequent negative lookups

Piecewise Linear-Regression

1. Breakdown data into subsets
2. Fit linear regression per subset



Student presentation

Bourbon paper importance

- Tackles scenarios with frequent writes
- Draws insights from workload characterization
- Cost-benefit filtering (waiting)

Discussion topics

1. How to support complex key types such as strings? How would the approach change?
2. What assumptions does the cost analyzer make? What are the limitations?